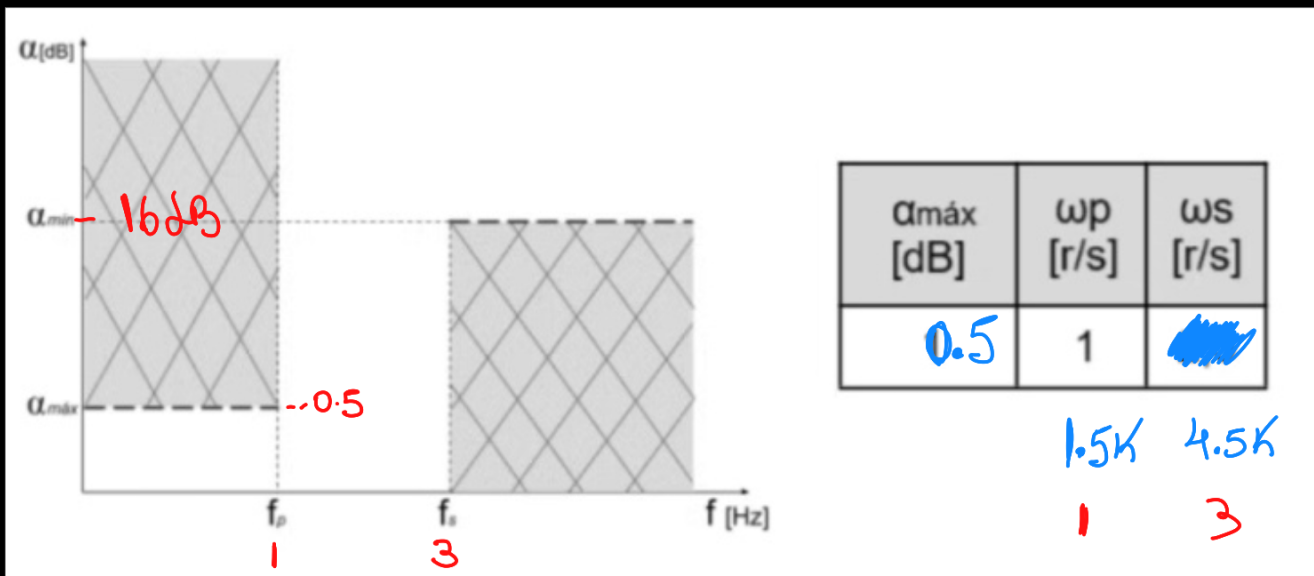




Usar un chebyshev

$$n = \frac{\cosh^{-1} \left(\left[\frac{\omega^{\alpha_{\min}/\omega} - 1}{\omega^{\alpha_{\max}/\omega} - 1} \right]^{1/2} \right)}{\cosh^{-1}(\omega_s)} \approx 2.02$$

$$n = 2$$

$$\xi^2 = \omega^{\alpha_{\max}/\omega} - 1 = 0.122$$

$$\xi = 0.3493$$

(Si se quiere
un filtro más
exigente $n=3$)

Se realiza el análisis con $n=2$
y $n=3$

Usando $n=2$

$$|T(j\omega)|^2 = \frac{1}{1 + \xi^2 C_n^2(\omega)} \quad ; \quad C_2(\omega) = 2\omega^2 - 1$$

$$|T(j\omega)|^2 \Big|_{\omega = \frac{s}{j}} = T(s)T(-s) = \frac{1}{1 + \xi^2 \left(2\left(\frac{s}{j}\right)^2 - 1 \right)^2}$$

$$T(s)T(-s) = \frac{1}{1 + \xi^2 (-2s^2 - 1)^2} = \frac{1}{1 + \xi^2 [4s^4 + 4s^2 + 1]}$$

$$T(s)T(-s) = \frac{\frac{1}{\xi^2 4}}{s^4 + s^2 + \frac{1 + \xi^2}{\xi^2 4}} = \frac{c}{s^2 + sa + b} \cdot \frac{c}{s^2 - sa + b}$$

$$c^2 = \frac{1}{\xi^2 4} \quad b^2 = \frac{1 + \xi^2}{\xi^2 4}$$

$$(2b - a^2) = 1$$

$$a^2 = 2b - 1$$

$$T(s) = \frac{c}{s^2 + s\alpha + b}$$

$$c = \frac{1}{\xi^2}$$

$$b = \sqrt{1 + \xi^2} \cdot c$$

$$\xi = 0.3493 \quad \omega_0 = \sqrt{b} = 1.231$$

$$Q = \frac{\sqrt{b}}{\alpha} = 0.863$$

$$\alpha = \sqrt{2b - 1}$$

$$c = 1.431$$

$$b = 1.5162$$

$$\alpha = 1.4256$$

$$K = \frac{c}{b} = 0.9438$$

Cheby $n=2$

$$T(s) = 0.944 \frac{\overbrace{\omega_0^2}^{1.5162}}{s^2 + s \underbrace{\frac{1.231}{0.863}}_{\frac{\omega_0}{Q}} + \underbrace{1.5162}_{\omega_0^2}}$$

Aukerberg - Mossberg

$$K = -\frac{R_3}{R_1}$$

$$\omega_0 = \frac{1}{R_3 C}$$

$$Q = \frac{R_2}{R_3}$$

$$R_3 = R_2 = 1$$

$$K = -\frac{1}{R_1}$$

$$\omega_0 = \frac{1}{C}$$

$$Q = R_2$$

$$U_{\text{Sando}} \quad n=3 \quad |T(j\omega)|^2 = \frac{1}{1 + \xi^2 C_n^2(\omega)} \quad ; \quad C_3 = 4\omega^3 - 3\omega$$

$$|T(j\omega)|^2 \Big|_{\omega = \frac{s}{j}} = T(s)T(-s) = \frac{1}{1 + \xi^2 \left(4\left(\frac{s}{j}\right)^3 - 3\frac{s}{j}\right)^2}$$

$$T(s)T(-s) = \frac{1}{1 + \xi^2 \left(-4\frac{s^3}{j} - 3\frac{s}{j}\right)^2}$$

$$T(s)T(-s) = \frac{1}{1 + \xi^2 \left[-16s^6 - 24s^4 - 9s^2\right]}$$

$$T(s)T(-s) = \frac{\frac{1}{16\xi^2}}{-s^6 - \frac{24s^4}{16} - \frac{9s^2}{16} + \frac{1}{\xi^2 16}}$$

$$\frac{d}{s^3 + cs^2 + bs + d} \cdot \frac{d}{-s^3 + cs^2 - bs + d} = \frac{\frac{1}{16\xi^2}}{-s^6 - \frac{24s^4}{16} - \frac{9s^2}{16} + \frac{1}{\xi^2 16}}$$

$$\rightarrow s^3 - s^3 = -s^6$$

$$\rightarrow s^3 \cdot s^2 - s^3 \cdot s^2 = 0$$

$$\rightarrow -bs^4 - bs^4 + c^2 s^4 \rightarrow -2b + c^2 = -\frac{24}{16} \quad (1)$$

$$\rightarrow a^2 = \frac{1}{\sum_j^2 16} = a = \frac{1}{\sum_j^4} \quad \text{; } d = a$$

$$\rightarrow -b^2 s^2 + 2ca s^2 \Rightarrow (2ca - b^2) = -\frac{a}{16} \quad (2)$$

$$C = \sqrt{2b - \frac{b}{4}} \quad (1) \quad C = \left(-\frac{a}{16} + b^2\right) \sum_j^2 \quad (2)$$

$$(1) = (2) \rightarrow b \begin{cases} \rightarrow 1.5349 \\ \rightarrow 0.75 \rightarrow \text{en } (1) \rightarrow c=0 \\ \therefore \text{Se descarta} \end{cases}$$

$$C = 1.2529 \quad a = 0.7157$$

Chebyshev $n=3$

$$T(s) = \frac{0.7157}{s^3 + 1.2529s^2 + 1.5349s + 0.7157}$$